MATH4010 FINAL PROJECT - GROUP C

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Google Colab Notebook Link: Group Project Notebook

1 QUESTION 1

We study two related families of balls:

$$A_n = \left\{ x \in \mathbb{R}^n : \left\| x - \frac{1}{2}\mathbf{1} \right\|_2 \le \frac{1}{2} \right\}, \qquad B_n = \{x \in \mathbb{R}^n : \|x\|_2 \le 1\}.$$

The set A_n is the largest ball inside the unit box $[0,1]^n$, while B_n is the unit ball centered at the origin. Since A_n has radius $1/2$ and B_n has radius 1, their volumes satisfy

$$\mathrm{Vol}(A_n) = 2^{-n}\,\mathrm{Vol}(B_n).$$

The purpose of this question is not only to compute areas and volumes, but also to understand how numerical estimation changes when dimension increases. We first study dimensions 2 and 3, where geometry is visible and sampling works well. Then we move to higher dimensions, where the ball becomes an extremely rare region inside its bounding box. This creates a natural story: simple methods work in low dimensions, fail in high dimensions, and motivate importance sampling.

1.1 Part 1.a: Two-Dimensional Case

In dimension 2, A_2 is a circle centered at $(1/2, 1/2)$ with radius $1/2$, and B_2 is the unit circle centered at the origin.

The exact area of a two-dimensional ball with radius r is

$$V_2(r) = \pi r^2.$$

Therefore,

$$\mathrm{Area}(A_2) = \pi\left(\frac{1}{2}\right)^2 = \frac{\pi}{4} \approx 0.785398,$$

and

$$\mathrm{Area}(B_2) = \pi(1)^2 = \pi \approx 3.141593.$$

Naive Monte Carlo Method

For a shape S inside a bounding box D, naive Monte Carlo estimates volume by sampling uniformly from D and counting how many samples fall inside S (**?**). If $X \sim \mathrm{Unif}(D)$, then

$$\mathrm{Vol}(S) = \mathrm{Vol}(D)\mathbb{P}(X \in S).$$

With N samples and N_{in} accepted points,

$$\widehat{\mathrm{Vol}}(S) = \mathrm{Vol}(D)\frac{N_{\text{in}}}{N}.$$

For A_n, the bounding box is $[0,1]^n$, whose volume is 1. For B_n, the bounding box is $[-1,1]^n$, whose volume is 2^n.

We use $N = 100$ and $N = 10000$. The smaller value $N = 100$ shows the rough behavior of the estimator, while $N = 10000$ shows the improvement after increasing the sample size. We do not focus on higher N in the low-dimensional baseline because the purpose is not only to reduce error, but to observe the convergence pattern and later compare it with the high-dimensional failure.

Metrics

We report four main metrics. The estimate is the computed area or volume. The number inside, N_{in}, shows how many sampled points fall inside the target ball. The acceptance rate is N_{in}/N, which measures how much of the bounding box is occupied by the ball. The absolute and relative errors are

$$\text{Abs. Err.} = |\widehat{V} - V|, \qquad \text{Rel. Err.} = \frac{|\widehat{V} - V|}{V} \times 100\%.$$

These metrics are useful because they show both numerical accuracy and sampling efficiency.

The two shapes have the same acceptance rate in dimension 2. This happens because A_2 is a scaled and shifted version of B_2, and its bounding box is scaled in the same way. Therefore, the fraction of the box occupied by the ball is unchanged.

Table 1: *Naive Monte Carlo results for A_2 and B_2.*

N	Shape	Exact	Estimate	Acc. Rate	Abs. Err.	Rel. Err.
100	A_2	0.7854	0.7600	76.00%	0.0254	3.23%
100	B_2	3.1416	3.0400	76.00%	0.1016	3.23%
10000	A_2	0.7854	0.7886	78.86%	0.0032	0.41%
10000	B_2	3.1416	3.1544	78.86%	0.0128	0.41%

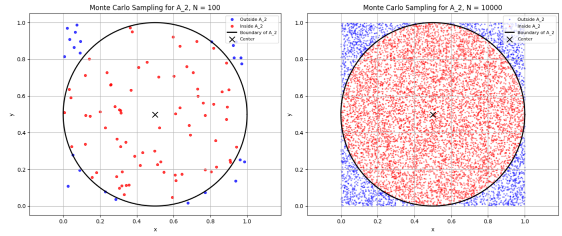

Figure 1: *Naive Monte Carlo visualization for A_2.*

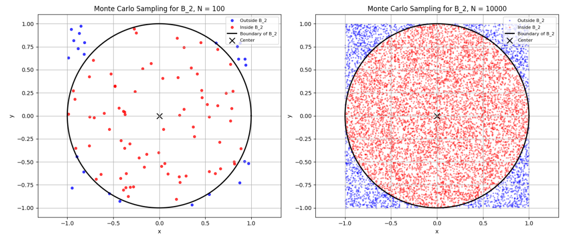

Figure 2: *Naive Monte Carlo visualization for B_2.*

1.2 Part 1.b: Three-Dimensional Case

In dimension 3, A_3 is a sphere centered at $(1/2, 1/2, 1/2)$ with radius $1/2$, and B_3 is the unit sphere centered at the origin.

The exact volume of a three-dimensional ball with radius r is

$$V_3(r) = \frac{4}{3}\pi r^3.$$

Thus,

$$\mathrm{Vol}(A_3) = \frac{4}{3}\pi\left(\frac{1}{2}\right)^3 = \frac{\pi}{6} \approx 0.523599,$$

and

$$\mathrm{Vol}(B_3) = \frac{4\pi}{3} \approx 4.188790.$$

We now apply the same naive Monte Carlo method. The comparison with dimension 2 reveals the first sign of dimensional difficulty.

Table 2: *Naive Monte Carlo results in dimensions 2 and 3.*

N	Shape	Exact	Estimate	Acc. Rate	Abs. Err.	Rel. Err.
100	A_2	0.7854	0.7600	76.00%	0.0254	3.23%
100	B_2	3.1416	3.0400	76.00%	0.1016	3.23%
100	A_3	0.5236	0.5000	50.00%	0.0236	4.51%
100	B_3	4.1888	4.0000	50.00%	0.1888	4.51%
10000	A_2	0.7854	0.7886	78.86%	0.0032	0.41%
10000	B_2	3.1416	3.1544	78.86%	0.0128	0.41%
10000	A_3	0.5236	0.5304	53.04%	0.0068	1.30%
10000	B_3	4.1888	4.2432	53.04%	0.0544	1.30%

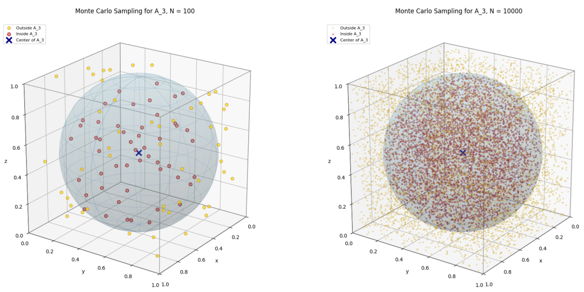

Figure 3: *Naive Monte Carlo visualization for A_3.*

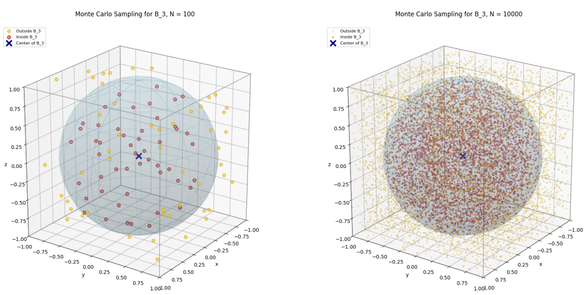

Figure 4: *Naive Monte Carlo visualization for B_3.*

The relative error decreases when N increases, but the acceptance rate drops from about 78.86% in dimension 2 to about 53.04% in dimension 3. This is the first warning sign: as dimension increases, the ball occupies a smaller fraction of the surrounding box.

1.3 Part 1.c: General Dimension and High-Dimensional Failure

For any $n \geq 1$, the volume of an n-dimensional ball with radius r is given by DLMF Eq. (5.19.4) (1):

$$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} r^n.$$

Therefore,

$$\mathrm{Vol}(B_n) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)},$$

and

$$\mathrm{Vol}(A_n) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} \left(\frac{1}{2}\right)^n = 2^{-n}\,\mathrm{Vol}(B_n).$$

This formula also explains why the problem becomes difficult. Although the ball volume is positive for every finite n, the fraction of the bounding box occupied by the ball decreases extremely fast as n increases. To see this transition before dimension 100, we test naive Monte Carlo with $N = 10000$ for dimensions 2, 5, 10, and 20.

Table 3: *Naive Monte Carlo performance as dimension increases, using $N = 10000$.*

Shape	Exact Volume	MC Estimate	Inside	Acc. Rate	Rel. Err.
A_2	7.8540×10^{-1}	7.8860×10^{-1}	7886	78.86%	0.41%
B_2	3.1416×10^{0}	3.1544×10^{0}	7886	78.86%	0.41%
A_5	1.6449×10^{-1}	1.6560×10^{-1}	1656	16.56%	0.67%
B_5	5.2638×10^{0}	5.2992×10^{0}	1656	16.56%	0.67%
A_{10}	2.4904×10^{-3}	2.5000×10^{-3}	25	0.25%	0.39%
B_{10}	2.5502×10^{0}	2.5600×10^{0}	25	0.25%	0.39%
A_{20}	2.4611×10^{-8}	0	0	0%	100%
B_{20}	2.5807×10^{-2}	0	0	0%	100%

This intermediate experiment shows the failure developing gradually. In dimension 2, the acceptance rate is high. In dimension 5, it drops to 16.56%. In dimension 10, only 25 out of 10000 samples are accepted, so the estimate becomes much more sensitive to randomness. By dimension 20, no sampled point is accepted, and the Monte Carlo estimate becomes zero.

This is the key insight for Part 1.c: the failure does not suddenly appear at dimension 100. It is already visible at moderate dimensions. As $n \to \infty$, the ball occupies an increasingly tiny fraction of the bounding box, so the acceptance probability of naive Monte Carlo goes to zero. Therefore, simply increasing N helps very slowly; the more important problem is that most samples are generated in regions that do not contribute to the volume.

For $n = 100$, the exact volumes are

$$\mathrm{Vol}(B_{100}) \approx 2.368202 \times 10^{-40}, \qquad \mathrm{Vol}(A_{100}) \approx 1.868182 \times 10^{-70}.$$

Table 4: *Exact volumes of A_{100} and B_{100}.*

Shape	Exact Volume	Log Volume
A_{100}	1.868182×10^{-70}	-160.556
B_{100}	2.368202×10^{-40}	-91.241

Although B_{100} has a larger volume than A_{100}, both are extremely small relative to their bounding boxes. For A_{100}, the bounding box has volume 1, so

$$p_A = \mathrm{Vol}(A_{100}) \approx 1.868182 \times 10^{-70}.$$

For B_{100}, the bounding box $[-1, 1]^{100}$ has volume 2^{100}, so

$$p_B = \frac{\mathrm{Vol}(B_{100})}{2^{100}} \approx 1.868182 \times 10^{-70}.$$

Thus, both problems have the same acceptance probability.

Even with $N = 100000$, the expected number of accepted points is

$$Np \approx 100000(1.868182 \times 10^{-70}) = 1.868182 \times 10^{-65}.$$

Even $N = 1000000$ gives only

$$Np \approx 1.868182 \times 10^{-64}.$$

Both values are essentially zero. To expect just one accepted point, we would need approximately

$$\frac{1}{p} \approx 5.3528 \times 10^{69}$$

samples, which is computationally impossible.

Table 5: *Naive Monte Carlo results for A_{100} and B_{100}.*

N	Shape	Exact	Estimate	Acc. Rate	Rel. Err.
100	A_{100}	1.87×10^{-70}	0	0%	100%
100	B_{100}	2.37×10^{-40}	0	0%	100%
10000	A_{100}	1.87×10^{-70}	0	0%	100%
10000	B_{100}	2.37×10^{-40}	0	0%	100%

The issue is not the formula for the exact volume. The issue is the sampling distribution. Uniform points from the full bounding box almost never land inside the target ball.

1.4 Exploration: Three Existing Methods

To check whether the failure is specific to naive Monte Carlo, we compare three methods: naive Monte Carlo, deterministic grid, and Sobol quasi-Monte Carlo.

Naive Monte Carlo uses independent random points. It is simple and flexible, but its accuracy depends strongly on how often samples hit the target region (?). The grid method uses evenly spaced deterministic points. It removes randomness, but it becomes expensive quickly because m points per coordinate gives m^n total points. Sobol quasi-Monte Carlo uses low-discrepancy points that cover the box more evenly than independent random samples (? ?). This usually improves low-dimensional performance.

For the final low-dimensional comparison, we report only $N = 10000$. The $N = 100$ case is useful for showing rough Monte Carlo behavior, but it is too noisy for a fair method comparison.

Table 6: *Comparison of naive MC, grid, and Sobol QMC in low dimensions using $N = 10000$.*

Shape	Method	Estimate	Acc. Rate	Abs. Err.	Rel. Err.
A_2	Naive MC	0.7886	78.86%	0.0032	0.41%
B_2	Naive MC	3.1544	78.86%	0.0128	0.41%
A_3	Naive MC	0.5304	53.04%	0.0068	1.30%
B_3	Naive MC	4.2432	53.04%	0.0544	1.30%
A_2	Grid	0.7668	76.68%	0.0186	2.37%
B_2	Grid	3.0672	76.68%	0.0744	2.37%
A_3	Grid	0.4485	44.85%	0.0751	14.34%
B_3	Grid	3.5883	44.85%	0.6005	14.34%
A_2	Sobol QMC	0.7857	78.57%	0.0003	0.04%
B_2	Sobol QMC	3.1428	78.57%	0.0012	0.04%
A_3	Sobol QMC	0.5219	52.19%	0.0017	0.32%
B_3	Sobol QMC	4.1752	52.19%	0.0136	0.32%

In low dimensions, Sobol QMC performs best. With $N = 10000$, it gives the smallest relative errors for both dimensions 2 and 3. However, the deeper pattern is that all three methods still explore the full bounding box. This is acceptable in low dimensions, where the ball occupies a visible part of the box, but it becomes ineffective in dimension 100, where almost every point in the box lies outside the ball.

Thus, the methods differ in how they place points, but they share the same weakness: they place points over the whole box instead of near the important region. This motivates the final method.

1.5 Final Method: Importance Sampling

Importance sampling changes the sampling distribution. Instead of drawing points uniformly from the full bounding box, it draws points from a proposal distribution concentrated near the target region (?). For a ball centered at c, we use a Gaussian proposal

$$q(x) = \mathcal{N}(c, \sigma^2 I).$$

The volume can be written as

$$\mathrm{Vol}(S) = \int \mathbf{1}_{\{x\in S\}}\, dx = \mathbb{E}_q\left[\frac{\mathbf{1}_{\{x\in S\}}}{q(x)}\right].$$

Thus,

$$\widehat{\mathrm{Vol}}(S) = \frac{1}{N}\sum_{i=1}^{N} \frac{\mathbf{1}_{\{x_i\in S\}}}{q(x_i)}, \qquad x_i \sim q.$$

For dimension 100, we choose

$$\sigma = \frac{r}{\sqrt{n}}.$$

This makes the typical squared distance from the center approximately

$$\mathbb{E}\|X - c\|^2 = n\sigma^2 = r^2.$$

Therefore, samples concentrate near the ball instead of being wasted far away.

For the importance sampling table, we add one more metric: standard error. It measures the sampling uncertainty of the weighted estimator. Smaller standard error means the weighted estimates are more stable across samples.

Table 7: *Importance sampling results for A_{100} and B_{100} with $N = 100000$.*

Shape	σ	Exact	IS Estimate	Std. Error	Acc. Rate	Rel. Err.
A_{100}	0.05	1.868182×10^{-70}	1.873099×10^{-70}	1.665×10^{-72}	52.01%	0.2632%
B_{100}	0.10	2.368202×10^{-40}	2.374435×10^{-40}	2.111×10^{-42}	52.01%	0.2632%

The improvement is significant. Naive Monte Carlo gives zero accepted points and 100% relative error for A_{100} and B_{100}. Importance sampling gives more than half of the samples inside the ball and relative error about 0.2632%. The method succeeds because it fixes the real issue: not the number of samples, but where the samples are placed.

1.6 Conclusion

The low-dimensional experiments show that naive Monte Carlo is a useful baseline, and Sobol QMC can outperform both naive Monte Carlo and the grid method when the dimension is small. However, all three methods share the same failure pattern: they sample across the full bounding box. As $n \to \infty$, the ball occupies an extremely small fraction of the box, so the probability of hitting the ball goes rapidly to zero.

This is why simply increasing N from 10000 to 100000 or 1000000 is not enough in dimension 100. The correct improvement is to change the sampling strategy. Importance sampling does this by sampling near the important region and weighting the samples properly. Among the tested methods, it is the most effective method for the high-dimensional case.

2 QUESTION 2

2.1 Estimate Z by direct calculations

The function e^{-x^4} is a symmetric function on $\mathbb{R}$ which is symmetric about the y-axis. As $|x|$ approaches to 0, it approaches to 0 extremely rapidly. Since $e^{-2^4} = e^{-16} \approx 1.12e^{-7}$, we will integrate the function from 0 to about 2 in practice.

Step 1: Sample the function. We divide $[0, 2]$ into 4 slices of width $\Delta x = 0.5$ and read off the heights:

x	$f(x) = e^{-x^4}$	Value
0.0	e^0	1.000
0.5	$e^{-0.0625}$	0.939
1.0	e^{-1}	0.368
1.5	$e^{-5.0625}$	0.006
2.0	e^{-16}	≈ 0

Step 2: Trapezoid rule (from 0 to 2). Each trapezoid's area $= \frac{\text{left points}+\text{right points}}{2} \times \Delta x$. Therefore, the sum of four slices approximates to:

$$0.5 \times \left(\frac{1.000+0.939}{2} + \frac{0.939+0.368}{2} + \frac{0.368+0.006}{2} + \frac{0.006+0}{2}\right)$$

Step 3: Add it up.

$$0.969 + 0.654 + 0.187 + 0.003 = 1.813$$

$$1.813 \times 0.5 = 0.906$$

Step 4: Double for symmetry. Since e^{-x^4} is symmetric around $x = 0$, the left half equals the right half:

$$Z \approx 0.906 \times 2 = \boxed{1.812}$$

This is remarkably close to the result calculated by scipy.integrate.quad.

2.2 Proposal Distributions

Three symmetric proposal distributions were tested, all satisfying $q(y|x) = q(x|y)$ so the Metropolis-Hastings acceptance ratio simplifies to $r = f(x')/f(x)$.

2.2.1 *Uniform Proposal*

$$x' = x + U(-\delta, \delta), \quad \delta = 1.$$

Symmetry holds because

$$q(y \mid x) = \begin{cases} \frac{1}{2\delta} & |y - x| \le \delta \\ 0 & \text{otherwise,} \end{cases}$$

and $q(x \mid y)$ is identical since $|y - x| = |x - y|$.

2.2.2 *Fixed-Step Random Walk Proposal*

$$x' = x + a \quad \text{or} \quad x' = x - a,$$

each with probability $\frac{1}{2}$.

Symmetry holds because

$$q(y \mid x) = \begin{cases} \frac{1}{2} & |y - x| = \frac{1}{2} \\ 0 & \text{otherwise,} \end{cases}$$

and $q(x \mid y)$ is identical since $|y - x| = |x - y|$.

2.2.3 *Normal Proposal*

$$x' = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 1.$$

Because $\mathcal{N}(0, \sigma^2)$ is symmetric, the probability of proposing a new state y from the current state x is equal to the probability of proposing x from y.

Symmetry holds because

$$q(y \mid x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(y-x)^2}{2\sigma^2}\right)$$

and $q(x \mid y)$ is identical since $|y - x| = |x - y|$.

2.2.4 Visualization

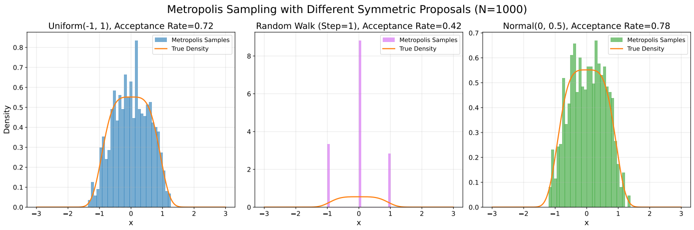

Figure 5: *Comparison of the three symmetric proposal distributions: uniform proposal, fixed-step random walk proposal, and normal proposal.*

2.3 Metropolis Algorithm

Starting from $x_0 = 0$, at each iteration:

1. Propose x' from the chosen symmetric distribution.
2. Compute the acceptance ratio. Since $f(x) = e^{-x^4}/Z$ and the proposal is symmetric:

$$r = \frac{e^{-(x')^4}/Z}{e^{-x^4}/Z} = e^{x^4-(x')^4}.$$

3. Accept x' with probability $\alpha = \min(1, r)$; otherwise keep x.

Repeat for $N = 1000$ samples.

2.4 Numerical Performance

Table 8: *Metropolis sampler performance ($N = 5000$).*

Proposal	Acc. Rate	$\bar{x}$	MAE	$\lvert E[X^4]_{\text{err}}\rvert$
Uniform	72.91%	0.0164	0.0164	0.015276
Rand. Walk	43.64%	−0.0032	0.0032	0.170871
Normal	78.78%	0.0074	0.0074	0.036198

2.5 Performance Discussion

Uniform Proposal. The uniform proposal achieved an acceptance rate of approximately 72.91%. Since the proposal distribution samples uniformly from the interval

$$(-1, 1),$$

every candidate step length inside the interval has equal probability of being proposed. This allows the Markov chain to explore the state space while maintaining a relatively high probability of accepting proposed samples.

The empirical distribution gradually converges toward the target density as the sample size increases.

Fixed-Step Random Walk. The fixed-step random walk proposal produced the lowest acceptance rate, approximately 43.64%. Since the proposal always moves by the same fixed distance, the chain cannot adapt its movement according to the local structure of the target distribution.

Although the chain still converges, the rigid proposal structure leads to weaker mixing behavior and the largest fourth-moment estimation error among the three methods.

Normal Proposal. The normal proposal achieved the highest acceptance rate, approximately 78.78%, outperforming both the uniform proposal (72.91%) and the fixed-step random walk proposal (43.64%).

This occurs because the normal distribution proposes candidate points near the current state more frequently than points far away. In contrast, the uniform proposal assigns equal probability to every step length inside the interval $(-1, 1)$.

Since the target density

$$f(x) = e^{-x^4}$$

decreases rapidly as $|x|$ increases, smaller proposal steps are more likely to remain inside high-probability regions and therefore have larger acceptance probabilities.

Consequently,

$$\min(1, \text{acceptance\_ratio}_{\text{normal}}) > \min(1, \text{acceptance\_ratio}_{\text{uniform}}),$$

which explains why the normal proposal achieves the largest overall acceptance rate.

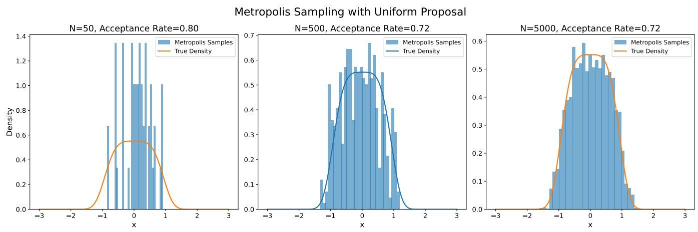

Figure 6: *Uniform proposal: histogram vs. true density for $N = 50, 500, 5000$.*

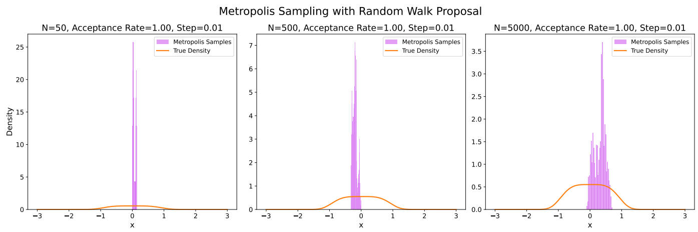

Figure 7: *Fixed-step random walk: histogram vs. true density for $N = 50, 500, 5000$.*

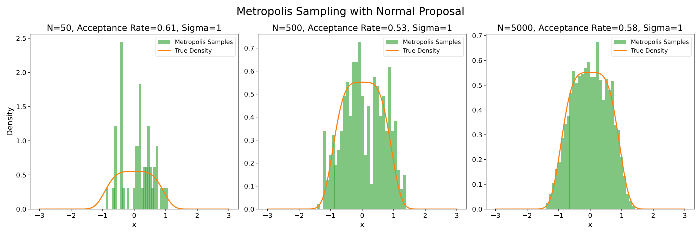

Figure 8: *Normal proposal: histogram vs. true density for $N = 50, 500, 5000$.*

2.6 Trace Plots & Convergence

Trace plots confirm that the uniform and normal proposals explore both sides of the symmetric target distribution effectively. The random walk proposal is noticeably more sluggish because of its fixed proposal step size.

Running means converge toward the theoretical expectation

$$E[X] = 0,$$

which follows from the symmetry of the target density:

$$\bar{x}_{\text{uniform}} \approx 0.0164, \quad \bar{x}_{\text{normal}} \approx 0.0074, \quad \bar{x}_{\text{rw}} \approx -0.0032.$$

The normal proposal produces the closest estimate to the theoretical mean, while the random walk proposal exhibits weaker convergence behavior in higher-order moments.

Autocorrelation analysis also indicates that the normal proposal decorrelates more rapidly than the fixed-step random walk, confirming stronger mixing efficiency for adaptive proposal structures.

2.7 Effect of Sample Size

Table 9: *Acceptance rates vs. sample size.*

N	Uniform (δ=1)	Rand. Walk	Normal (σ=1)
50	0.7600	0.4286	0.7959
500	0.7420	0.4399	0.7836
5000	0.7291	0.4364	0.7878

As the sample size increases, the empirical distributions produced by the uniform and normal proposals become smoother and converge more closely toward the target density.

The normal proposal consistently maintains the highest acceptance rate because proposed samples closer to the current state occur with higher probability. Meanwhile, the fixed-step random walk proposal consistently produces the lowest acceptance rate and weakest mixing behavior because of its rigid proposal structure.

2.8 Suggested Improvements

1. **Burn-in**: discard the first few hundred samples to reduce dependence on the initial state $x_0 = 0$.
2. **Tune proposal parameters**: optimize δ, proposal step size, and σ to improve convergence and mixing efficiency.
3. **Diagnostics**: compute effective sample size (ESS) and integrated autocorrelation time for more rigorous convergence analysis.

2.9 Conclusion (Q2)

The Metropolis algorithm successfully sampled from

$$f(x) = \frac{e^{-x^4}}{Z},$$

with

$$Z \approx 1.8128049541109543.$$

Among the three proposal distributions, the normal proposal achieved the highest acceptance rate (78.78%), outperforming the uniform proposal (72.91%) and the fixed-step random walk proposal (43.64%).

The normal proposal performs well because candidate points near the current state are proposed more frequently. Since the target density decreases rapidly as $|x|$ increases, smaller proposal steps are more likely to remain in regions with high probability density and therefore have larger acceptance probabilities.

In contrast, the uniform proposal assigns equal probability to every step length inside $(-1, 1)$, while the fixed-step random walk proposal always proposes moves of identical magnitude. Consequently, the normal proposal produces larger values of

$$\min(1, r),$$

leading to a larger overall acceptance rate.

The experiments also demonstrate that acceptance rate alone does not completely determine sampling quality. Efficient exploration and strong mixing behavior are equally important for accurately approximating the target distribution.

3 QUESTION 3

3.1 Data

We recorded 20 audio clips: 10 ambient-noise clips and 10 music clips. The recordings were made using a Samsung Galaxy Z Fold 4 with the Voice Recorder app, which saved the raw recordings in .m4a format. The clips were edited and converted using Audacity, and the final audio files were exported in .wav format. All final clips had a duration of 5 s. Ambient noise included room noise, fan/AC noise, traffic, and street rumble; music included short melodies, chords, singing, and songs played through a phone speaker. Before entropy estimation, each clip was peak-normalized to $\max_n |x_n| = 1$ so that all clips share the same dynamic range $[-1, 1]$. This makes the common bin width comparable across clips; the normalization shifts $\widehat{H}$ and H_{Gauss} by the same constant, so the gap $H_{\text{Gauss}} - \widehat{H}$ is preserved.

3.2 Compute four numbers per clip

For every peak-normalized clip, I used the same bin width $\Delta = 0.005$. Let $\widehat{p}_b = n_b/N$ be the empirical probability in bin b, and let K be the number of bins covering the clip's sample range. The four reported quantities are

$$\widehat{H}_Q = -\sum_b \widehat{p}_b \log_2 \widehat{p}_b, \qquad \widehat{H} = \widehat{H}_Q + \log_2 \Delta,$$

$$H_{\max} = \log_2 K, \qquad H_{\text{Gauss}} = \tfrac{1}{2}\log_2(2\pi e \widehat{\sigma}^2).$$

Here $\widehat{H}_Q$ is the discrete entropy of the quantized samples, $\widehat{H}$ is the histogram-based differential entropy estimate, $H_{\max}$ is the uniform-bin upper bound, and H_{Gauss} is the maximum-entropy reference for a continuous variable with the same variance. (2; 3).

Table 10: *Mean $\pm$ standard deviation of entropy quantities across 10 clips per class. All values are in bits except $\widehat{\sigma}^2$.*

Quantity	Music	Ambient noise
$\widehat{H}_Q$	7.08 ± 0.44	6.16 ± 1.20
$\widehat{H}$	-0.56 ± 0.44	-1.48 ± 1.20
$H_{\max}$	8.59 ± 0.05	8.51 ± 0.12
H_{Gauss}	-0.42 ± 0.40	-1.15 ± 1.01
$H_{\text{Gauss}} - \widehat{H}$	0.14 ± 0.14	0.33 ± 0.32
$\widehat{\sigma}^2$	0.04 ± 0.02	0.03 ± 0.07

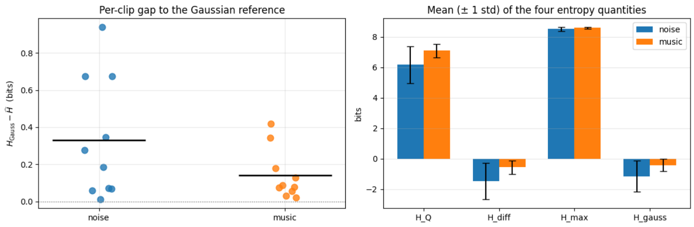

Figure 9: *Comparison of entropy-based statistics for ambient-noise and music clips.* ***Left:*** *Per-clip Gaussian entropy gap, $H_{\text{Gauss}} - \hat{H}$, for each audio clip, with horizontal black lines indicating the mean gap for each class. Smaller values indicate amplitude distributions that more closely match a Gaussian distribution with the same variance. Music clips generally exhibit smaller Gaussian entropy gaps than ambient-noise clips.* ***Right:*** *Mean ± 1 standard deviation of the four entropy quantities (H_Q, H_{diff}, $H_{\max}$, and H_{gauss}) for the two audio classes.*

3.3 Discussion

Table 10 and Figure 9 show that music has the smaller typical Gaussian gap: 0.14 bits on average, compared with 0.33 bits for ambient noise. The medians tell the same story: 0.08 bits for music and 0.23 bits for noise. Therefore, in these recordings, the music clips are closer to Gaussian in the time-domain amplitude sense. The result is not saying that music is truly Gaussian; rather, its empirical amplitude distribution is closer to the matched-variance Gaussian than the recorded ambient-noise clips.

This interpretation follows directly from the maximum-entropy-given-variance result: among all continuous distributions with fixed variance σ^2, the Gaussian distribution has the largest differential entropy, with value $\frac{1}{2}\log_2(2\pi e \sigma^2)$ (3). Thus $H_{\text{Gauss}} - \widehat{H}$ measures how much entropy is lost relative to the Gaussian with the same variance, and equality would occur only for a Gaussian density. The ambient-noise clips have a larger and more variable gap because many of them are mostly quiet with occasional transients, giving a sharply peaked, sparse, non-Gaussian amplitude distribution. The music clips contain more sustained tones and chords, so their sample amplitudes are spread more continuously across the normalized range.

The sanity checks also agree with theory. For every clip, $\widehat{H}_Q \leq H_{\max}$, because the uniform distribution over bins maximizes discrete entropy. Also, $\widehat{H} \leq H_{\text{Gauss}}$ for every clip, so all Gaussian gaps are non-negative.

3.4 Further Exploration

First, a bin-width sensitivity analysis recomputes all entropy quantities for $\Delta \in \{2^{-15}, \ldots, 2^{-8}\}$. This test examines whether the observed difference between music and ambient noise is genuinely structural or merely an artifact of the histogram discretization. Histogram-based entropy estimators are known to depend strongly on bin width: excessively fine bins increase estimator variance and sparse-bin effects, whereas excessively coarse bins introduce bias by oversmoothing the distribution (4; 5). Since the baseline choice $\Delta = 0.005$ is somewhat arbitrary, evaluating a wide range of bin widths provides a robustness check for the main conclusion. Figure 10 shows that the estimated entropy $\widehat{H}$ decreases substantially as the bins become finer, particularly for ambient noise, reflecting the increasing sensitivity of histogram estimators at high resolutions. In contrast, the Gaussian entropy estimate H_{Gauss} remains comparatively stable because it depends only on the sample variance. Most importantly, the Gaussian entropy gap $H_{\text{Gauss}} - \widehat{H}$ remains positive for both classes across all tested bin widths, and the ambient-noise gap consistently exceeds the music gap. Although the magnitude of the gap changes with Δ, the qualitative ordering is preserved throughout the experiment. This indicates that the conclusion—ambient noise exhibits a larger deviation from Gaussianity than music under this metric—is not caused by any particular histogram resolution choice.

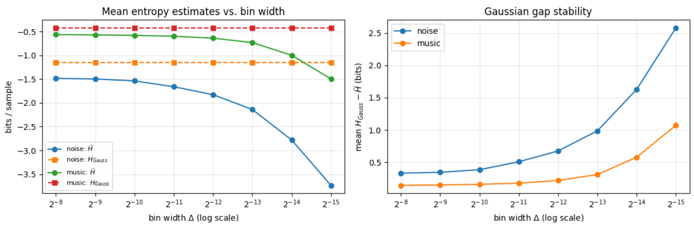

Figure 10: *Sensitivity of entropy estimates and Gaussian entropy gaps to histogram bin width. Although the estimated entropy $\widehat{H}$ varies substantially as the bin width changes, the Gaussian entropy gap $H_{\text{Gauss}} - \widehat{H}$ remains consistently larger for ambient noise than for music across all tested resolutions. This indicates that the qualitative conclusion is robust to the histogram discretization choice.*

Second, histogram and Q–Q plots help explain the mechanism behind the entropy-gap differences by showing how each signal deviates from a Gaussian distribution with matched variance. For a fixed variance, the Gaussian distribution achieves the maximum possible entropy (6); therefore, larger values of $H_{\text{Gauss}} - \widehat{H}$ indicate stronger departures from Gaussian structure. The representative ambient-noise clips show strong concentration near zero together with uneven or heavy tails, whereas the music clips more closely follow the matched Gaussian envelope in both the histograms and Q–Q plots. This visual evidence is consistent with the quantitative entropy-gap results and provides an interpretable explanation for why ambient noise produces systematically larger gaps. Figure 11 illustrates representative clips near the median gap value for each class.

Third, a bootstrap analysis provides an uncertainty estimate for the observed Gaussian entropy gaps. The bootstrap is useful here because the sample size is small (10 clips per class), making analytic variance estimates less reliable; instead, repeated resampling approximates the sampling distribution directly (7). The estimated mean gap for ambient noise is 0.33 with a 95% confidence interval of $[0.15, 0.53]$, while music has a mean gap of 0.14 with interval $[0.07, 0.23]$. The estimated noise-minus-music difference is therefore 0.19 bits with confidence interval $[-0.01, 0.41]$. Although the interval slightly overlaps zero, likely due to the limited number of clips, the effect size remains practically meaningful with Cohen's $d = 0.76$, indicating a moderate-to-large separation between the two classes.

4 References

[1] "*NIST Digital Library of Mathematical Functions*," https://dlmf.nist.gov/, Release 1.2.6 of 2026-03-15, f. W. J. Olver, A. B. Olde Daalhuis, D. W. Lozier, B. I. Schneider, R. F. Boisvert, C. W. Clark, B. R. Miller, B. V. Saunders, H. S. Cohl, and M. A. McClain, eds. [Online]. Available: https://dlmf.nist.gov/

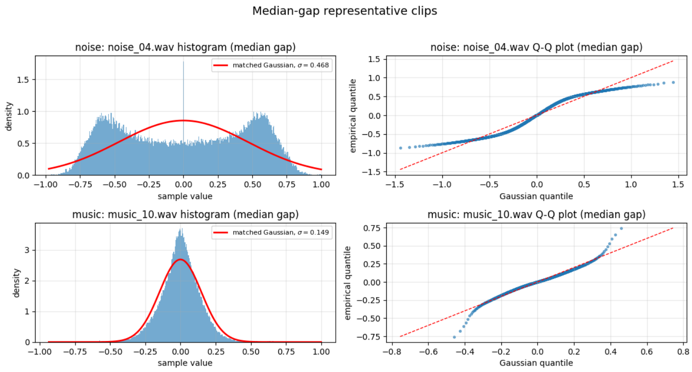

Figure 11: *Representative histogram and Q–Q plots for clips near the median Gaussian entropy gap in each class. Ambient-noise clips exhibit stronger deviations from the matched Gaussian distribution, including concentration near zero and heavier or uneven tails, while music clips more closely follow the Gaussian reference. These visual differences help explain the larger Gaussian entropy gaps observed for ambient noise.*

[2] C. E. Shannon, "A mathematical theory of communication," *Bell System Technical Journal*, vol. 27, no. 3, pp. 379–423, 1948.

[3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.

[4] D. Freedman and P. Diaconis, "On the histogram as a density estimator: L2 theory," *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, vol. 57, no. 4, pp. 453–476, 1981.

[5] L. Paninski, "Estimation of entropy and mutual information," *Neural Computation*, vol. 15, no. 6, pp. 1191–1253, 2003.

[6] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[7] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

5 Contribution Declaration

Q	#	Sub-task	Leader	Member	QC
1	1	Visualize A_2, B_2, A_3, B_3	Duy Anh	N/A	Dinh Hieu
	2	General formula: exact volumes (small & large cases)	Thai An	N/A	
	3	Implement algorithm; report estimates, errors, metrics; compare	Anh Trung	Thai An	
	4	Challenges; discuss $N \to \infty$; suggest improvements	Duy Anh	Tue Anh	
2	5	Estimate normalization factor Z	Nhat Linh	N/A	Bao Tien
	6	2 distributions: Metropolis, N=1000, visualize, report	Bao Tien	N/A	
	7	2 distributions: Metropolis, N=1000, visualize, report	Nhat Minh	N/A	
	8	Discuss performance across proposal distributions	Duy Anh	N/A	
3	9	Data: 10 clips noise + 10 clips music	Dai Nghia	Nhat Linh	Duy Anh
	10	Code: discrete/differential/Gaussian entropy; mean & std	Dinh Hieu	N/A	
	11	Gap $H_{\text{Gauss}} - \widehat{H}$; compare classes; max-entropy result	Dinh Hieu	Duc Thanh	
–	12	Slide	Duc Thanh	Thai An, Tue Anh	
	13	Report	Nhat Linh	N/A	
	14	Poster	Tue Anh	N/A	N/A

Color code: **Red = Leader** **Blue = Member** **Green = QC**